Review Session 1

Wold Decomposition, Linear Filters, and ARMA Processes

Sangmyung Ha

Fall 2026

Throughout these notes, stationarity means covariance stationarity, all processes are indexed by $t \in \mathbb{Z}$, and all random variables are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Processes are centered unless stated otherwise. Infinite random-variable series are interpreted as limits in L^2. White noise means a mean-zero process with a constant covariance matrix and zero cross-covariances at distinct dates; independence is not required.

1 Multivariate Wold Decomposition

1.1 Scalar Hilbert spaces and linear prediction

We work in the scalar Hilbert space

$$\mathcal{H} = L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}), \qquad \langle U, V \rangle = \mathbb{E}(UV), \qquad \|U\|_2 = \{\mathbb{E}(U^2)\}^{1/2}.$$

Elements of $\mathcal{H}$ are measurable, square-integrable real random variables, with random variables that are equal almost surely identified. For a vector $Y = (Y_1, \ldots, Y_n)'$, we use the convenient notation $\|Y\|_2^2 = \sum_{i=1}^n \|Y_i\|_2^2 = \mathbb{E}(Y'Y)$, but every projection below is a scalar projection applied separately to each coordinate.

Let $X_t = (X_{1,t}, \ldots, X_{n,t})'$ be a mean-zero stationary process with $\mathbb{E}\|X_t\|^2 < \infty$. Define

$$\mathcal{H}_t^X = \overline{\text{span}}\{X_{i,s} : 1 \le i \le n,\ s \le t\}, \tag{1.1}$$

$$\mathcal{H}^X = \overline{\text{span}}\{X_{i,s} : 1 \le i \le n,\ s \in \mathbb{Z}\}, \qquad \mathcal{H}_{-\infty}^X = \bigcap_{t \in \mathbb{Z}} \mathcal{H}_t^X. \tag{1.2}$$

Here $\overline{\text{span}}$ denotes the L^2 closure of the set of finite real linear combinations. Membership in $\mathcal{H}_t^X$ therefore means that a scalar random variable can be approximated in L^2 by finite linear combinations of the coordinates of current and past observations.

Let P_t be the orthogonal projection onto $\mathcal{H}_t^X$, and write

$$P_t X_s = (P_t X_{1,s}, \ldots, P_t X_{n,s})'.$$

The one-step linear innovation is

$$\varepsilon_t = X_t - P_{t-1} X_t. \tag{1.3}$$

For every coordinate i, the projection theorem gives

$$\varepsilon_{i,t} \in \mathcal{H}_t^X, \qquad \mathbb{E}(\varepsilon_{i,t} Y) = 0 \quad \text{for every } Y \in \mathcal{H}_{t-1}^X.$$

The predictor $P_{t-1}X_t$ is the best linear predictor; it need not coincide with the conditional expectation given the past.

1.2 Multivariate Wold Decomposition

Theorem 1.1 (Multivariate Wold decomposition with full-rank innovations)**.** *Let $\{X_t\}_{t\in\mathbb{Z}}$ be a mean-zero, stationary, $\mathbb{R}^n$-valued process with finite second moments. Define its one-step innovations by* (1.3)*, using the scalar prediction spaces* (1.1)*, and assume that*

$$\Sigma_\varepsilon := \mathbb{E}(\varepsilon_0\varepsilon_0') \succ 0. \tag{1.4}$$

Then $\{\varepsilon_t\}$ is vector white noise with covariance Σ_ε. There are unique matrices $\{\Theta_k\}_{k\geq 0}$, relative to these innovations, such that

$$X_t = \sum_{k=0}^{\infty} \Theta_k \varepsilon_{t-k} + D_t, \tag{1.5}$$

where

$$\Theta_0 = I_n, \qquad \Theta_k = \mathbb{E}(X_t\varepsilon_{t-k}')\Sigma_\varepsilon^{-1}, \qquad \sum_{k=0}^{\infty} \operatorname{tr}(\Theta_k\Sigma_\varepsilon\Theta_k') < \infty. \tag{1.6}$$

The matrices Θ_k do not depend on t, and the series in (1.5) *converges in L^2. Each coordinate of D_t belongs to $\mathcal{H}^X_{-\infty}$, and*

$$\mathbb{E}(D_t\varepsilon_s') = 0 \qquad \textit{for all } s,t \in \mathbb{Z}.$$

The two components in (1.5) *are jointly stationary and orthogonal at all leads and lags. If the process is purely nondeterministic, meaning $\mathcal{H}^X_{-\infty} = \{0\}$, then $D_t = 0$ for all t.*

Proof. Step 1: Stationarity of the innovations. On finite linear combinations of the coordinates of X, define a forward shift U by

$$U\left(\sum_{(i,s)\in F} c_{i,s}X_{i,s}\right) = \sum_{(i,s)\in F} c_{i,s}X_{i,s+1},$$

where F is a finite set of coordinate–date pairs. Covariance stationarity implies that this map preserves inner products. It is therefore well defined on almost-sure equivalence classes and extends by continuity to an onto isometry of $\mathcal{H}^X$. The backward shift is its inverse.

Since $U\mathcal{H}^X_t = \mathcal{H}^X_{t+1}$, preservation of orthogonality implies

$$UP_tY = P_{t+1}UY \qquad \text{for every } Y \in \mathcal{H}^X.$$

Consequently,

$$U\varepsilon_{i,t} = U(X_{i,t} - P_{t-1}X_{i,t}) = X_{i,t+1} - P_tX_{i,t+1} = \varepsilon_{i,t+1}.$$

Thus X and ε are jointly covariance stationary. In particular,

$$\mathbb{E}(\varepsilon_t\varepsilon_t') = \Sigma_\varepsilon \qquad \text{for every } t,$$

and $\mathbb{E}(X_t\varepsilon_{t-k}')$ depends only on k.

For $s < t$, every coordinate $\varepsilon_{j,s}$ belongs to $\mathcal{H}^X_s \subset \mathcal{H}^X_{t-1}$. Hence

$$\mathbb{E}(\varepsilon_{i,t}\varepsilon_{j,s}) = 0 \qquad \text{for all } i,j,$$

which gives $\mathbb{E}(\varepsilon_t \varepsilon_s') = 0$. The innovations have mean zero because the prediction spaces are generated by mean-zero random variables. They are therefore vector white noise.

Step 2: The one-step orthogonal decomposition. Define the finite-dimensional innovation space

$$\mathcal{I}_t = \operatorname{span}\{\varepsilon_{1,t}, \dots, \varepsilon_{n,t}\}.$$

We claim that

$$\mathcal{H}_t^X = \mathcal{H}_{t-1}^X \oplus \mathcal{I}_t. \tag{1.7}$$

To see this, $\mathcal{H}_t^X$ is generated by $\mathcal{H}_{t-1}^X$ and the coordinates of X_t. For each i,

$$X_{i,t} = P_{t-1} X_{i,t} + \varepsilon_{i,t},$$

with $P_{t-1} X_{i,t} \in \mathcal{H}_{t-1}^X$ and $\varepsilon_{i,t} \in \mathcal{I}_t$. Conversely, $\varepsilon_{i,t} = X_{i,t} - P_{t-1} X_{i,t} \in \mathcal{H}_t^X$. Thus adding the coordinates of X_t to the past is equivalent to adding $\mathcal{I}_t$. Moreover, $\mathcal{I}_t \perp \mathcal{H}_{t-1}^X$, and their sum is closed because it is the orthogonal sum of two closed subspaces. This proves (1.7).

Let

$$\mathcal{E}_t = \overline{\operatorname{span}}\{\varepsilon_{i,s} : 1 \le i \le n,\ s \le t\} \subset \mathcal{H}_t^X.$$

If $Y \in \mathcal{H}_t^X$ is orthogonal to $\mathcal{E}_t$, then (1.7) first implies $Y \in \mathcal{H}_{t-1}^X$, because $Y \perp \mathcal{I}_t$. Since also $Y \perp \mathcal{I}_{t-1}$, it follows that $Y \in \mathcal{H}_{t-2}^X$. Repeating gives $Y \in \mathcal{H}_{-\infty}^X$.

Conversely, if $Y \in \mathcal{H}_{-\infty}^X$, then $Y \in \mathcal{H}_{s-1}^X$ for every s, so Y is orthogonal to every innovation coordinate. Therefore

$$\mathcal{H}_t^X = \mathcal{H}_{-\infty}^X \oplus \mathcal{E}_t. \tag{1.8}$$

Step 3: Projection coefficients and convergence. The coordinatewise projection of X_t onto $\mathcal{I}_{t-k}$ can be written as $\Theta_k \varepsilon_{t-k}$. Its scalar orthogonality conditions, one for each pair of coordinates, are equivalent to

$$\mathbb{E}\{(X_t - \Theta_k \varepsilon_{t-k}) \varepsilon_{t-k}'\} = 0.$$

Since $\Sigma_\varepsilon \succ 0$, these conditions uniquely determine

$$\Theta_k = \mathbb{E}(X_t \varepsilon_{t-k}') \Sigma_\varepsilon^{-1}.$$

Step 1 shows that these coefficients do not depend on t. Also,

$$\mathbb{E}(X_t \varepsilon_t') = \mathbb{E}\{(P_{t-1} X_t + \varepsilon_t) \varepsilon_t'\} = \Sigma_\varepsilon,$$

so $\Theta_0 = I_n$.

The spaces $\mathcal{I}_t, \mathcal{I}_{t-1}, \dots$ are mutually orthogonal. Therefore

$$S_t^{(K)} = \sum_{k=0}^{K} \Theta_k \varepsilon_{t-k}$$

is the coordinatewise projection of X_t onto $\mathcal{I}_t \oplus \cdots \oplus \mathcal{I}_{t-K}$. Pythagoras gives

$$\sum_{k=0}^{K} \operatorname{tr}(\Theta_k \Sigma_\varepsilon \Theta_k') = \|S_t^{(K)}\|_2^2 \le \|X_t\|_2^2. \tag{1.9}$$

The nonnegative series is consequently summable. For $M > K$,

$$\|S_t^{(M)} - S_t^{(K)}\|_2^2 = \sum_{k=K+1}^{M} \operatorname{tr}(\Theta_k \Sigma_\varepsilon \Theta_k') \longrightarrow 0$$

as $K, M \to \infty$. Completeness of L^2 gives a limit $S_t = \sum_{k\geq 0} \Theta_k \varepsilon_{t-k}$.

Each coordinate of S_t belongs to $\mathcal{E}_t$. For every fixed k, the residual $X_t - S_t^{(K)}$ is coordinatewise orthogonal to $\mathcal{I}_{t-k}$ whenever $K \geq k$. Taking limits shows that $X_t - S_t$ is coordinatewise orthogonal to $\mathcal{E}_t$. Thus S_t is the coordinatewise projection of X_t onto $\mathcal{E}_t$.

Step 4: The remote-past component and uniqueness. Set $D_t = X_t - S_t$. By (1.8), each coordinate of D_t belongs to $\mathcal{H}_{-\infty}^X$. In fact, $D_t = P_{-\infty} X_t$, where $P_{-\infty}$ denotes coordinatewise projection onto $\mathcal{H}_{-\infty}^X$.

For any s, t, the coordinates of D_t belong to $\mathcal{H}_{s-1}^X$, so $\mathbb{E}(D_t \varepsilon_s') = 0$. Taking L^2 limits in the series defining S_s also gives $\mathbb{E}(D_t S_s') = 0$ for every s, t.

The shift U maps $\mathcal{H}_{-\infty}^X$ onto itself and hence commutes with $P_{-\infty}$ on $\mathcal{H}^X$. Thus $UD_{i,t} = D_{i,t+1}$ and $US_{i,t} = S_{i,t+1}$, which proves joint covariance stationarity of the two components.

The remote-past component is unique by uniqueness of orthogonal projection. Taking cross-covariances of (1.5) with ε_{t-k} then uniquely determines every Θ_k, because Σ_ε is nonsingular. Finally, if $\mathcal{H}_{-\infty}^X = \{0\}$, then $D_t = 0$ for every t. □

Remark (What the theorem does and does not imply). Positive definiteness gives

$$\lambda_{\min}(\Sigma_\varepsilon)\|\Theta_k\|_F^2 \leq \operatorname{tr}(\Theta_k \Sigma_\varepsilon \Theta_k') \leq \lambda_{\max}(\Sigma_\varepsilon)\|\Theta_k\|_F^2.$$

Thus the Wold coefficients are square summable. The theorem does not assert absolute summability. Also, (1.5) gives the useful prediction formula

$$P_{t-1} X_t = D_t + \sum_{k=1}^{\infty} \Theta_k \varepsilon_{t-k}.$$

The innovation is exactly the part of X_t that cannot be predicted linearly from its past.

Remark (The full-rank assumption is a genuine restriction). Stationarity alone does not imply (1.4). For the process in Question 1 of Problem Set 1, write $X_t = (u_t, u_t + 0.5u_{t-1})'$ with $\{u_t\} \sim \mathrm{WN}(0, \sigma^2)$ and $\sigma^2 > 0$. Its one-step innovation is $(u_t, u_t)'$, with covariance

$$\sigma^2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

which is singular. The full-rank version of the theorem stated here does not apply to that example.

Example 1.2 (A deterministic component can be random)**.** Let $X_t = Z + u_t$, where Z is a mean-zero, square-integrable random variable orthogonal to $\{u_t\} \sim \mathrm{WN}(0, \sigma^2)$ with $\sigma^2 > 0$. For any fixed r,

$$\frac{1}{N} \sum_{j=0}^{N-1} X_{r-j} = Z + \frac{1}{N} \sum_{j=0}^{N-1} u_{r-j} \longrightarrow Z \qquad \text{in } L^2.$$

Hence $Z \in \mathcal{H}_r^X$ for every r, so $Z \in \mathcal{H}_{-\infty}^X$. The innovations are u_t and the Wold decomposition is $X_t = u_t + Z$. Here "deterministic" means linearly predictable from the arbitrarily remote past, not necessarily nonrandom.

2 Causality and Invertibility of MA and AR Processes

In this section, ε_t denotes a specified driving white noise. It is not automatically the one-step innovation of X_t. Unless explicitly stated otherwise, its covariance is positive definite in the vector case, and its variance is $\sigma^2 > 0$ in the scalar case.

2.1 Linear processes and terminology

Consider the two-sided VMA representation

$$X_t = \sum_{j=-\infty}^{\infty} \Theta_j \varepsilon_{t-j}, \qquad \sum_{j=-\infty}^{\infty} \|\Theta_j\|_F^2 < \infty, \tag{2.1}$$

where $X_t, \varepsilon_t \in \mathbb{R}^n$ and $\Theta_j \in \mathbb{R}^{n\times n}$. Orthogonality across dates implies convergence in L^2. In particular,

$$\mathbb{E}(X_t X_t') = \sum_{j=-\infty}^{\infty} \Theta_j \Sigma_\varepsilon \Theta_j'.$$

More generally, with $\Gamma_X(h) = \mathbb{E}(X_t X_{t-h}')$,

$$\Gamma_X(h) = \sum_{j=-\infty}^{\infty} \Theta_{j+h} \Sigma_\varepsilon \Theta_j', \qquad \Gamma_X(-h) = \Gamma_X(h)'. \tag{2.2}$$

The covariance series is absolutely convergent entrywise by the Cauchy–Schwarz inequality and square summability. Since it depends only on h, the process is covariance stationary.

Causality relative to the driving noise. The representation (2.1) is causal when $\Theta_j = 0$ for all $j < 0$, so that

$$X_t = \sum_{j=0}^{\infty} \Theta_j \varepsilon_{t-j}.$$

Under $\Sigma_\varepsilon \succ 0$, this is equivalent to

$$\mathbb{E}(\varepsilon_{t+k} X_t') = 0 \qquad \text{for all } k > 0,$$

because

$$\mathbb{E}(\varepsilon_{t+k} X_t') = \Sigma_\varepsilon \Theta_{-k}'.$$

Thus a causal representation uses current and past, but not future, driving noise.

Invertibility of an MA representation. An MA representation is invertible if there are matrices $\{\Pi_j\}_{j\geq 0}$ such that

$$\varepsilon_t = \sum_{j=0}^{\infty} \Pi_j X_{t-j}, \qquad \sum_{j=0}^{\infty} \|\Pi_j\|_F < \infty. \tag{2.3}$$

The driving noise can then be recovered through a one-sided, absolutely summable filter of the observations. This implies $\varepsilon_{i,t} \in \mathcal{H}_t^X$ for every coordinate i. The converse need not hold: membership in a closed linear span is weaker than the existence of an absolutely summable inverse filter.

Invertibility of an AR process For a VAR(p) equation

$$X_t = \sum_{j=1}^{p} A_j X_{t-j} + \varepsilon_t,$$

the lecture notes use *invertibility of the AR process* to mean that the AR filter can be inverted to obtain

$$X_t = \sum_{j=0}^{\infty} \Psi_j \varepsilon_{t-j}, \qquad \sum_{j=0}^{\infty} \|\Psi_j\|_F < \infty. \tag{2.4}$$

This is a causal, absolutely summable MA representation. Notice the different directions: AR-invertibility expresses X_t in terms of the driving noise, whereas MA-invertibility expresses the driving noise in terms of X_t.

Remark (Closed-span recoverability is weaker). Consider $X_t = u_t - u_{t-1}$ with $\{u_t\} \sim \mathrm{WN}(0, \sigma^2)$ and $\sigma^2 > 0$. The finite predictors

$$Z_{N,t} = \sum_{j=0}^{N-1} \left(1 - \frac{j}{N}\right) X_{t-j} = u_t - \frac{1}{N} \sum_{j=1}^{N} u_{t-j}$$

satisfy $\mathbb{E}|Z_{N,t} - u_t|^2 = \sigma^2/N$. Thus $u_t \in \mathcal{H}_t^X$. However, a one-sided inverse filter for $1 - L$ must have every coefficient equal to one, so it cannot be absolutely summable. The finite predictors above change their coefficients with N; they are not partial sums of an absolutely summable inverse filter.

2.2 MA models

A finite-order vector MA model is

$$X_t = \sum_{j=0}^{q} \Theta_j \varepsilon_{t-j}. \tag{2.5}$$

It is causal relative to the specified noise and covariance stationary for every fixed collection of finite coefficient matrices. Stationarity does not require MA-invertibility. From (2.2),

$$\Gamma_X(h) = \sum_{j=0}^{q-h} \Theta_{j+h} \Sigma_\varepsilon \Theta_j' \quad (0 \le h \le q), \qquad \Gamma_X(h) = 0 \quad (h > q).$$

Negative lags are obtained by transposition.

For a scalar MA(1),

$$X_t = \varepsilon_t + \theta \varepsilon_{t-1},$$

the inverse filter is

$$\varepsilon_t = \sum_{j=0}^{\infty} (-\theta)^j X_{t-j}$$

when $|\theta| < 1$. The coefficients are absolutely summable, and the root of $1 + \theta z$ lies strictly outside the unit circle when $\theta \neq 0$. The case $\theta = 0$ is trivially invertible. The general MA root condition is developed in Section 3.3.

Remark (Driving noise versus innovations). For a causal, MA-invertible representation with $\Theta_0 = I_n$, the driving noise coincides with the one-step innovation. Causality makes ε_t orthogonal to the past observations, while invertibility ensures that each coordinate of $\sum_{j\geq 1} \Theta_j \varepsilon_{t-j}$ belongs to $\mathcal{H}_{t-1}^X$. Thus

$$P_{t-1}X_t = \sum_{j=1}^{\infty} \Theta_j \varepsilon_{t-j}, \qquad X_t - P_{t-1}X_t = \varepsilon_t.$$

A qualification about timing. For any integer k, the process $\widetilde{X}_t = \sum_{j=0}^{q} \Theta_j \varepsilon_{t-k-j}$ has the same autocovariance function as (2.5). Thus autocovariances alone do not identify the time labels attached to latent driving shocks. Choosing $k = 0$ fixes this arbitrary shift, but whether a shock has a contemporaneous effect depends on its loading in Θ_0. This timing issue should not be confused with MA-invertibility or with the definition of the one-step innovation.

2.3 AR models: the AR(1) example

An AR equation does not by itself imply stationarity. We ask whether it admits a stationary solution and, if it does, whether that solution is causal relative to the specified driving noise. Consider the scalar model

$$X_t = aX_{t-1} + \varepsilon_t, \qquad \{\varepsilon_t\} \sim \mathrm{WN}(0, \sigma^2), \qquad \sigma^2 > 0. \tag{2.6}$$

Case 1: $|a| < 1$**.** The series

$$X_t = \sum_{j=0}^{\infty} a^j \varepsilon_{t-j} \tag{2.7}$$

converges in L^2, since for $M > N$,

$$\mathbb{E}\left| \sum_{j=N}^{M-1} a^j \varepsilon_{t-j} \right|^2 = \sigma^2 \sum_{j=N}^{M-1} |a|^{2j} \longrightarrow 0.$$

Its limit is stationary and satisfies (2.6), as can be checked by subtracting aX_{t-1} from (2.7).

To establish uniqueness among stationary solutions, iterate the recursion for any such solution:

$$X_t = \sum_{j=0}^{N-1} a^j \varepsilon_{t-j} + a^N X_{t-N}.$$

Stationarity gives

$$\|a^N X_{t-N}\|_2 = |a|^N \|X_0\|_2 \longrightarrow 0.$$

Hence every stationary solution equals (2.7). It is AR-invertible because $\sum_{j\geq 0} |a|^j < \infty$. Notice that the remainder argument uses the uniform second-moment bound supplied by stationarity, not merely square integrability at each separate date.

Case 2: $|a| > 1$**.** Rewrite the recursion forward as

$$X_t = a^{-1}X_{t+1} - a^{-1}\varepsilon_{t+1}.$$

Repeated substitution gives

$$X_t = -\sum_{j=1}^{N} a^{-j}\varepsilon_{t+j} + a^{-N}X_{t+N}.$$

For any stationary solution, $\|a^{-N}X_{t+N}\|_2 = |a|^{-N}\|X_0\|_2 \to 0$. Therefore the only possible stationary solution is

$$X_t = -\sum_{j=1}^{\infty} a^{-j}\varepsilon_{t+j}. \tag{2.8}$$

This series indeed converges in L^2, since $\sum_{j\geq 1} |a|^{-2j} < \infty$, and its limit is stationary and satisfies (2.6).

The solution is not causal relative to this driving noise. In particular,

$$\mathbb{E}(\varepsilon_{t+1}X_t) = -\frac{\sigma^2}{a} \neq 0.$$

Consequently the AR equation is not invertible in the one-sided sense used in the course, even though it has a stationary, two-sided-filter solution. Furthermore,

$$\mathbb{E}(\varepsilon_t X_{t-1}) = -\frac{\sigma^2}{a} \neq 0,$$

so the driving noise is not the one-step innovation of X. There is no conflict with Wold's theorem, which uses innovations defined by projection, not an arbitrary driving white noise.

Case 3: $|a| = 1$**.** There is no finite-variance stationary solution under $\sigma^2 > 0$. Indeed, since $a \in \mathbb{R}$, this case means $a = 1$ or $a = -1$, and iteration would imply

$$\mathbb{E}|X_t - a^N X_{t-N}|^2 = \mathbb{E}\left|\sum_{j=0}^{N-1} a^j \varepsilon_{t-j}\right|^2 = N\sigma^2.$$

But stationarity and the triangle inequality imply

$$\mathbb{E}|X_t - a^N X_{t-N}|^2 \leq 4\mathbb{E}|X_0|^2,$$

which contradicts $N\sigma^2 \to \infty$.

3 How to Check Invertibility for General AR and MA Processes

3.1 Taylor and Laurent expansions

We use two facts from complex analysis.

1. If f is analytic in the disk $|z - z_0| < R$, then its Taylor expansion is

$$f(z) = \sum_{j=0}^{\infty} \frac{f^{(j)}(z_0)}{j!}(z - z_0)^j \qquad (|z - z_0| < R).$$

2. If f is analytic in the annulus $r < |z - z_0| < R$, then it has a Laurent expansion

$$f(z) = \sum_{j=-\infty}^{\infty} b_j(z - z_0)^j \qquad (r < |z - z_0| < R).$$

The Laurent coefficients are unique *on the specified annulus*. A function can have different Laurent expansions on different annuli. These series converge absolutely and uniformly on compact subannuli.

There is an important distinction between the following two domains:

$$\{z : |z| = 1\} \quad \text{(unit circle)}, \qquad \{z : |z| \le 1\} \quad \text{(closed unit disk)}.$$

If f is analytic on an open neighborhood of the closed unit disk, its Taylor expansion about zero is valid on a disk of radius greater than one. If f is analytic only on an open neighborhood of the unit circle, there is an annulus $r < |z| < R$ with $r < 1 < R$ on which it has a Laurent expansion; negative powers may be needed.

For example, for $a \neq 0$,

$$\frac{1}{1 - az} = \sum_{j=0}^{\infty} a^j z^j \qquad \text{when } |az| < 1,$$

$$\frac{1}{1 - az} = -\sum_{j=1}^{\infty} a^{-j} z^{-j} \qquad \text{when } |az| > 1.$$

These two expansions correspond to the causal and noncausal AR(1) solutions. If $|a| = 1$, the pole lies on the unit circle, so neither domain contains a neighborhood of the entire unit circle.

Lemma 3.1 (Absolute summability of Laurent coefficients)**.** *Suppose f is analytic in an annulus $r < |z| < R$ with $r < 1 < R$, and write its Laurent expansion as $f(z) = \sum_{j\in\mathbb{Z}} b_j z^j$. Then*

$$\sum_{j=-\infty}^{\infty} |b_j| < \infty.$$

Proof. Choose r_0, R_0 such that $r < r_0 < 1 < R_0 < R$. Convergence of the nonnegative-power part at $z = R_0$ implies that $b_j R_0^j \to 0$ as $j \to \infty$. Hence there is a finite constant C_+ such that

$$|b_j| \le C_+ R_0^{-j} \qquad (j \ge 0).$$

Similarly, convergence of the negative-power part at $z = r_0$ implies that $b_{-j} r_0^{-j} \to 0$, so there is a finite constant C_- such that

$$|b_{-j}| \le C_- r_0^j \qquad (j \ge 1).$$

Both bounds are geometrically summable, giving

$$\sum_{j\in\mathbb{Z}} |b_j| \le \frac{C_+}{1-R_0^{-1}} + \frac{C_- r_0}{1-r_0} < \infty.$$

□

If f is analytic on a neighborhood of the closed unit disk, the same argument applies to its Taylor coefficients, and there are no negative-power terms. Thus analyticity on a neighborhood of the *circle* produces an absolutely summable two-sided filter, whereas analyticity on a neighborhood of the *closed disk* produces an absolutely summable one-sided filter.

3.2 AR(p): scalar polynomials and lag operators

Consider a scalar AR(p) equation

$$X_t = \sum_{j=1}^{p} \alpha_j X_{t-j} + \varepsilon_t.$$

For a stationary process, the lag operator L is first defined on finite linear combinations by $LX_t = X_{t-1}$ and then extended continuously to $\mathcal{H}^X$. Covariance stationarity makes L an onto isometry. In particular,

$$\|L^j Y\|_2 = \|Y\|_2 \qquad \text{for every } Y \in \mathcal{H}^X \text{ and every } j \in \mathbb{Z}.$$

Negative powers are leads: $L^{-j}X_t = X_{t+j}$ for $j > 0$.

The AR equation becomes

$$\alpha(L)X_t = \varepsilon_t, \qquad \alpha(L) = I - \sum_{j=1}^{p} \alpha_j L^j.$$

We associate with this operator the scalar polynomial

$$\alpha(z) = 1 - \sum_{j=1}^{p} \alpha_j z^j.$$

The polynomial $\alpha(z)$ is a function of a complex number; $\alpha(L)$ is an operator on random variables. A convergent coefficient expansion connects the two.

Why an absolutely summable filter is well defined. Let $\sum_{j\in\mathbb{Z}} |b_j| < \infty$, with real coefficients, and define

$$B_N(L) = \sum_{|j|\le N} b_j L^j.$$

For $M > N$,

$$\|B_M(L) - B_N(L)\|_{\text{op}} \le \sum_{N<|j|\le M} |b_j| \longrightarrow 0,$$

where $\|\cdot\|_{\mathrm{op}}$ is the operator norm induced by $\|\cdot\|_2$. Thus $B_N(L)$ converges in operator norm to a bounded operator $B(L)$ on $\mathcal{H}^X$. For every $Y \in \mathcal{H}^X$,

$$B(L)Y = \sum_{j\in\mathbb{Z}} b_j L^j Y, \qquad \|B(L)Y\|_2 \le \left(\sum_{j\in\mathbb{Z}} |b_j|\right) \|Y\|_2.$$

In particular, the filtered series belongs to the same closed space and satisfies

$$\mathbb{E}\left|\sum_{j\in\mathbb{Z}} b_j X_{t-j}\right|^2 \le \mathbb{E}|X_0|^2 \left(\sum_{j\in\mathbb{Z}} |b_j|\right)^2. \tag{3.1}$$

This argument establishes convergence before taking expectations of an infinite series.

A two-sided inverse when there are no unit-circle roots. Suppose

$$\alpha(z) \neq 0 \qquad \text{for every } |z| = 1. \tag{3.2}$$

Because α is a finite polynomial, $1/\alpha(z)$ is analytic on an annulus containing the unit circle. Write

$$\frac{1}{\alpha(z)} = \sum_{j\in\mathbb{Z}} b_j z^j.$$

Lemma 3.1 gives $\sum_j |b_j| < \infty$. Since α has real coefficients, uniqueness of the Laurent expansion implies that the b_j are real as well.

The identity $\alpha(z)\sum_j b_j z^j = 1$ and uniqueness of Laurent coefficients give

$$b_j - \sum_{\ell=1}^{p} \alpha_\ell b_{j-\ell} = \begin{cases} 1, & j = 0, \\ 0, & j \neq 0. \end{cases}$$

Absolute convergence permits multiplication and regrouping of the corresponding operator series. Hence

$$\alpha(L)B(L) = B(L)\alpha(L) = I, \qquad B(L) = \sum_{j\in\mathbb{Z}} b_j L^j.$$

This proves that $B(L)$ is actually the inverse, not merely a well-defined filter. Every stationary solution must therefore satisfy

$$X_t = \alpha(L)^{-1}\varepsilon_t = \sum_{j\in\mathbb{Z}} b_j \varepsilon_{t-j}. \tag{3.3}$$

Conversely, starting with white noise, this series constructs a stationary solution: it converges in L^2, and the coefficient identities verify the AR equation. Thus (3.2) guarantees a unique stationary solution, possibly involving future driving noise.

A one-sided inverse when all roots are outside the closed disk. Suppose the stronger condition

$$\boxed{\alpha(z) \neq 0 \qquad \text{for every } |z| \leq 1} \tag{3.4}$$

holds. Then $1/\alpha(z)$ is analytic on a neighborhood of the closed unit disk and has a Taylor expansion

$$\frac{1}{\alpha(z)} = \sum_{j=0}^{\infty} \psi_j z^j, \qquad \sum_{j=0}^{\infty} |\psi_j| < \infty.$$

Consequently,

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

so the AR process is invertible in the course's terminology.

The condition is also necessary for such an absolutely summable one-sided representation when $\sigma^2 > 0$. Indeed, substituting the representation into the AR equation and using orthogonality of the white noise forces the convolution coefficients to equal those of the identity filter. Thus, with $\psi(z) = \sum_{j \geq 0} \psi_j z^j$,

$$\alpha(z)\psi(z) = 1 \qquad \text{for every } |z| \leq 1.$$

Absolute summability makes $\psi(z)$ continuous and finite on the closed unit disk, so α cannot vanish there.

Examples. For AR(1), $\alpha(z) = 1 - az$ has its finite root at $z = 1/a$ when $a \neq 0$. The one-sided condition is $|a| < 1$; $a = 0$ is trivially included. For AR(2),

$$\alpha(z) = 1 - \alpha_1 z - \alpha_2 z^2,$$

and every finite root must have modulus strictly greater than one. Roots on the unit circle are excluded. Analyticity merely on $|z| < 1$, without a neighborhood of the closed disk, is not enough.

3.3 MA(q) and ARMA root conditions

For a scalar MA(q), write

$$X_t = \beta(L)\varepsilon_t, \qquad \beta(z) = 1 + \sum_{j=1}^{q} \beta_j z^j.$$

If

$$\boxed{\beta(z) \neq 0 \qquad \text{for every } |z| \leq 1,} \tag{3.5}$$

then

$$\frac{1}{\beta(z)} = \sum_{j=0}^{\infty} \pi_j z^j, \qquad \sum_{j=0}^{\infty} |\pi_j| < \infty,$$

and the operator argument gives

$$\varepsilon_t = \beta(L)^{-1} X_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}.$$

Conversely, an absolutely summable one-sided inverse implies $\beta(z)\pi(z) = 1$ on the closed unit disk by the same white-noise coefficient argument as for AR(p). Hence (3.5) is necessary and sufficient for MA-invertibility. It is not a condition for stationarity of a finite MA process.

For a scalar ARMA(p, q),

$$\alpha(L)X_t = \beta(L)\varepsilon_t,$$

assume $\alpha(0) = \beta(0) = 1$ and that α and β have no common roots, as in the lecture notes. The two distinct inversions are

$$X_t = \frac{\beta(L)}{\alpha(L)}\varepsilon_t \qquad \text{if } \alpha(z) \neq 0 \text{ for } |z| \leq 1,$$

$$\varepsilon_t = \frac{\alpha(L)}{\beta(L)}X_t \qquad \text{if } \beta(z) \neq 0 \text{ for } |z| \leq 1.$$

Each fraction denotes the corresponding absolutely summable one-sided filter. The first gives a causal MA representation; the second recovers the driving noise from the observations of a stationary solution. The lecture notes' use of "invertible ARMA" refers to inversion of its AR component, so it is useful to state explicitly which direction is intended.

3.4 A different approach: companion matrices

For the scalar AR(2), define

$$Y_t = \begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix}, \qquad A = \begin{pmatrix} \alpha_1 & \alpha_2 \\ 1 & 0 \end{pmatrix}, \qquad e_t = \begin{pmatrix} \varepsilon_t \\ 0 \end{pmatrix}.$$

Then

$$Y_t = AY_{t-1} + e_t,$$

and repeated substitution gives

$$Y_t = \sum_{j=0}^{N-1} A^j e_{t-j} + A^N Y_{t-N}.$$

If every eigenvalue of A has modulus strictly less than one, then $\rho(A) < 1$, where $\rho(A)$ is the spectral radius. For any r with $\rho(A) < r < 1$, there is a finite constant C_r such that

$$\|A^j\|_{\mathrm{op}} \leq C_r r^j \qquad (j \geq 0).$$

This also holds when A is not diagonalizable: the polynomial factors arising from Jordan blocks are dominated by a geometric bound with ratio r.

It follows that $\sum_{j\geq 0} \|A^j\|_{\mathrm{op}} < \infty$ and $\sum_{j\geq 0} \|A^j\|_{\mathrm{op}}^2 < \infty$. Since the e_t are orthogonal across dates,

$$\mathbb{E}\left\| \sum_{j=N}^{M-1} A^j e_{t-j} \right\|^2 \leq \mathbb{E}\|e_0\|^2 \sum_{j=N}^{M-1} \|A^j\|_{\mathrm{op}}^2 \longrightarrow 0.$$

Thus $\sum_{j\geq 0} A^j e_{t-j}$ converges in L^2 and defines a stationary solution. For any stationary solution, the remainder satisfies

$$\|A^N Y_{t-N}\|_2 \leq C_r r^N \|Y_0\|_2 \longrightarrow 0.$$

Consequently,

$$Y_t = \sum_{j=0}^{\infty} A^j e_{t-j}. \tag{3.6}$$

The covariance of e_t is singular, which does not affect this argument. The full-rank innovation assumption in Theorem 1.1 is not needed for the companion-matrix calculation.

Which roots are inside and which are outside? The companion eigenvalues solve

$$\det(\lambda I_2 - A) = \lambda^2 - \alpha_1\lambda - \alpha_2 = 0.$$

On the other hand,

$$\det(I_2 - zA) = 1 - \alpha_1 z - \alpha_2 z^2 = \alpha(z).$$

Thus a nonzero companion eigenvalue λ corresponds to the finite AR-polynomial root $z = 1/\lambda$. Therefore

$$\boxed{\rho(A) < 1 \quad \Longleftrightarrow \quad \alpha(z) \neq 0 \text{ for every } |z| \leq 1.} \tag{3.7}$$

The *companion eigenvalues* must be strictly *inside* the unit circle; the *AR-polynomial roots* must be strictly *outside*. A zero companion eigenvalue corresponds to no finite root of the AR polynomial and creates no difficulty.

3.4.1 The AR(2) coefficient region

For real α_1, α_2, the equivalent coefficient restrictions are

$$\boxed{\alpha_2 > -1, \qquad \alpha_1 + \alpha_2 < 1, \qquad \alpha_2 - \alpha_1 < 1.} \tag{3.8}$$

Here is a direct verification. Let λ_1, λ_2 be the roots of $q(\lambda) = \lambda^2 - \alpha_1\lambda - \alpha_2$. If both have modulus below one, then

$$|\alpha_2| = |\lambda_1\lambda_2| < 1, \qquad q(1) = 1 - \alpha_1 - \alpha_2 > 0, \qquad q(-1) = 1 + \alpha_1 - \alpha_2 > 0.$$

The positivity statements hold for real roots in $(-1, 1)$ and for a complex-conjugate pair, where the products are squared moduli. These inequalities imply (3.8).

Conversely, (3.8) implies $|\alpha_2| < 1$ and $q(1), q(-1) > 0$. If the roots are nonreal, they are conjugates and satisfy $|\lambda_1|^2 = |\lambda_2|^2 = -\alpha_2 < 1$. If a real root were greater than or equal to one, positivity of $q(1)$ would force both roots to exceed one, contradicting $|\lambda_1\lambda_2| < 1$. The same argument using $q(-1)$ excludes a root less than or equal to minus one. Hence both roots lie in $(-1, 1)$.

Equivalently, the admissible region is $-1 < \alpha_2 < 1 - |\alpha_1|$. It is the interior of the triangle with vertices $(-2, -1)$, $(2, -1)$, and $(0, 1)$; its boundary is excluded.

3.4.2 AR(2) autocovariances

Under (3.8), let $\gamma(h) = \mathbb{E}(X_t X_{t-h})$. Since the stationary solution is causal, ε_t is orthogonal to past observations. Multiplying the AR(2) equation by X_{t-h} and taking expectations gives

$$\begin{aligned}
\gamma(1) &= \alpha_1\gamma(0) + \alpha_2\gamma(1), \\
\gamma(2) &= \alpha_1\gamma(1) + \alpha_2\gamma(0), \\
\gamma(0) &= \alpha_1\gamma(1) + \alpha_2\gamma(2) + \sigma^2.
\end{aligned}$$

Solving yields

$$\gamma(0) = \frac{\sigma^2(1-\alpha_2)}{(1+\alpha_2)\{(1-\alpha_2)^2 - \alpha_1^2\}}, \qquad \gamma(1) = \frac{\alpha_1}{1-\alpha_2}\gamma(0). \tag{3.9}$$

The remaining autocovariances are obtained recursively from

$$\gamma(h) = \alpha_1\gamma(h-1) + \alpha_2\gamma(h-2) \qquad (h \geq 2), \qquad \gamma(-h) = \gamma(h).$$

The autocorrelation function is $\gamma(h)/\gamma(0)$. This connects the root conditions to the autocovariance calculations in Problem Set 1.

3.4.3 Extension to VAR(p)

For $X_t \in \mathbb{R}^n$ satisfying $X_t = A_1X_{t-1} + \cdots + A_pX_{t-p} + \varepsilon_t$, use the block companion matrix

$$\mathsf{A} = \begin{pmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}.$$

For $p = 1$, simply take $\mathsf{A} = A_1$. With

$$\mathcal{A}(z) = I_n - \sum_{j=1}^{p} A_j z^j,$$

block elimination gives

$$\det(I_{np} - z\mathsf{A}) = \det \mathcal{A}(z).$$

Therefore the causal, absolutely summable VAR representation is obtained under

$$\boxed{\rho(\mathsf{A}) < 1 \quad \Longleftrightarrow \quad \det \mathcal{A}(z) \neq 0 \text{ for every } |z| \leq 1.} \tag{3.10}$$

The resulting MA coefficients satisfy

$$\Psi_0 = I_n, \qquad \Psi_j = \sum_{\ell=1}^{\min(p,j)} A_\ell \Psi_{j-\ell} \quad (j \geq 1), \qquad \sum_{j=0}^{\infty} \|\Psi_j\|_F < \infty.$$

The root condition concerns the *full matrix polynomial*, or equivalently the block companion matrix, not the eigenvalues of the individual lag matrices considered separately.

For a square vector MA polynomial $\Theta(z) = \Theta_0 + \cdots + \Theta_q z^q$, the analogous sufficient and necessary condition for an absolutely summable one-sided inverse, with full-rank driving noise, is

$$\det \Theta(z) \neq 0 \qquad \text{for every } |z| \leq 1.$$

This includes nonsingularity of Θ_0. The inverse matrix function is analytic on a neighborhood of the closed unit disk, so its Taylor coefficients supply the inverse filter.

Course references

These notes accompany *Stationary Processes in the Time Domain*, Sections 2.1–2.4, and Problem Set 1 (Fall 2026). The term "invertible AR process" follows the convention in Section 2.3 of the lecture notes.